

Ethnus

Some common math problems:

1. $487 + 296$
 $+3 \quad +4 \quad \Rightarrow \quad (+7)$

$490 + 300 =$

790
-7
783

2. $356 + 478$
 $+4 \quad +2 \quad \Rightarrow \quad (+6)$

$360 + 480 =$

840
-6
834

3. $\underline{23} + \underline{47} + \underline{18} + \underline{12} + \underline{77} + \underline{53}$

$23 + 77 = 100$

$18 + 12 = 30$

$$47 + 53 = 100$$

$$100 + 30 + 100 = 230$$

4. $1000 - 467$

$$10 - 7 = 3$$

$$9 - 6 = 3$$

$$9 - 4 = 5$$

$$\boxed{533}$$

5. $824 - 298$

$+2$

$$\begin{array}{r} 824 \\ - 300 \\ \hline 524 \end{array}$$

$$\begin{array}{r} 524 \\ + 2 \\ \hline 526 \end{array}$$

6. $5 \frac{1}{3}$ of 5 + $373 \frac{1}{3}$ of 1 + ? =

$$5^2 \times 4^2$$

$$\frac{16}{3} \times 5 + \frac{1120}{3} \times 1 + x = 25 \times 16$$

$$\frac{80}{3} + \frac{1120}{3} + x = 400$$

$$\frac{1200}{3} + x = 400$$

$$400 + x = 400$$

$$x = 400 - 400$$

$$x = 0$$

7. $9543 - (2341 + 4325) + 1208 = ?$

$$\underline{9543} - 6666 + \underline{1208}$$

$$10751 - 6666$$

$$4085$$

8. $88888 + 8888 + 888 + 88 + 8 = ?$

$$8(11111 + 1111 + 111 + 11 + 1)$$

$$8(12345)$$

$$98760$$

9. $13 \frac{3}{4} + 17 \frac{2}{7} + 31 \frac{1}{4} + 15 \frac{5}{7} + 12 \frac{2}{3} = ?$

$$(13 + 17 + 31 + 15 + 12) + \left(\frac{3}{4} + \frac{2}{7} + \frac{1}{4} + \frac{5}{7} + \frac{2}{3} \right)$$

$$88 + \left(1\frac{1}{4} + \frac{7}{7} + \frac{2}{3} \right)$$

$$88 + (1 + 1 + \frac{2}{3})$$

$$88 + (2 + \frac{2}{3})$$

$$(88 + 2) + \frac{2}{3}$$

$$\boxed{90 \frac{2}{3}}$$

10. $0.9999 + 0.999 + 0.99 + 0.9 = ?$

$$9 (0.1111 + 0.111 + 0.11 + 0.1)$$

$$9 (0.54321)$$

$$\boxed{3.8889}$$

$$11. \quad 49 \frac{7}{8} + 25 \frac{15}{16} - 19 \frac{3}{4} + 30 \frac{1}{8} - 14 \frac{7}{16}$$

$$(49 + 25 - 19 + 30 - 14) + \left(\frac{7}{8} + \frac{15}{16} - \frac{3}{4} + \frac{1}{8} - \frac{7}{16} \right)$$

$$71 + \left(\frac{8}{8} + \frac{8}{16} - \frac{3}{4} \right)$$

$$71 + \left(1 + \frac{1}{2} - \frac{3}{4} \right)$$

$$71 + \left(\frac{3}{2} - \frac{3}{4} \right)$$

$$71 + \left(\frac{6-3}{4} \right)$$

$$71 + \frac{3}{4}$$

$$\boxed{71 \frac{3}{4}}$$

$$12. \quad \frac{1}{11} - \frac{2}{11} + \frac{3}{11} - \frac{4}{11} + \frac{5}{11} = ?$$

$$\boxed{\frac{3}{11}}$$

13. $\underline{2502} - \underline{1001} + \underline{499} - \underline{201} + \underline{702} = ?$

$$2502 + 499 + 702 = 3703$$

$$1001 + 201 = 1202$$

$$3703 - 1202 = 2501$$

14. $\underline{1999} - \underline{998} + \underline{501} - \underline{302} + \underline{300}$

$$1999 + 501 + 300 = 2800$$

$$998 + 302 = 1300$$

$$2800 - 1300 = 1500$$

15. $\underline{1499} + \underline{501} - \underline{999} + \underline{301} - \underline{2}$

$$1499 + 501 + 301 = 2301$$

$$999 + 2 = 1001$$

$$2301 - 1001 = 1300$$

Squaring the numbers :

From 1 to 75 :

$$\begin{array}{l} (46)^2 \nearrow 25 \xrightarrow{\text{diff}} \textcircled{21} \\ \searrow 50 \xrightarrow{(\text{diff})^2} 4^2 = \textcircled{16} \end{array}$$

Ans : 2116

From 76 to 1000 :

$$\begin{array}{l} (79)^2 \nearrow 100 \xrightarrow{\text{diff}} \textcircled{21} \\ \searrow \begin{array}{r} 79 \\ - 21 \\ \hline 58 \end{array} \end{array}$$

$(21)^2 = 441$

①
$$\begin{array}{r} 58 \\ 441 \\ \hline 6241 \end{array}$$

Ans : 6241

Square rooting the number:

Squares

unit number

1 9

1

2 8

2

3 7

9

4 6

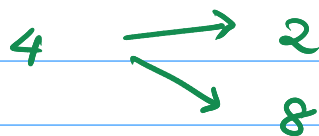
6

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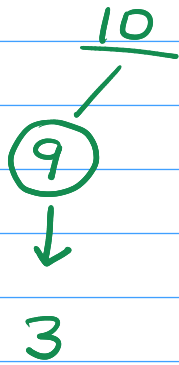
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$$\sqrt{1024}$$

- i. Identify the unit number and guess the squared value number.



- ii. Find the nearest square number for the first two numbers.



iii. Multiply the number with the next number.

$$(3) \times 4 = 12$$

iv. Compare the First two number of Square root Value With the multiplied Value:

If big. unit value big

$$12 > 10$$

If small. unit value small.

$$\hookrightarrow (2)$$

Ans : 3 2

16. $(1.6)^2 \div (0.8)^2 = [(2.4)^2 \div (0.4)^2] - x$
 $- ?$

$$\left(\frac{16}{10}\right)^2 \times \left(\frac{10}{8}\right)^2 = \left[\left(\frac{24}{10}\right)^2 \times \left(\frac{10}{4}\right)^2\right] - x$$

$$\frac{16 \times 16}{8 \times 8} = \frac{24 \times 24}{4 \times 4} - x$$

$$4 = 36 - x$$

$$x = 36 - 4$$

$$x = 32$$

17. $\sqrt{37249} \times \sqrt{2809} - (87)^2 = x^2 + (48)^2$
 $- 5$

$$\sqrt{37249} = 193$$

$$\begin{array}{r} 193 \\ 1 \overline{) 37249} \\ \underline{1} \\ 272 \\ \underline{261} \\ 1149 \\ \underline{1149} \\ 0 \end{array}$$

$$\sqrt{2809} \begin{array}{l} \textcircled{3} \\ \swarrow \\ 7 \end{array}$$

$$\textcircled{5} \times 6$$

$$30 > \underline{28} \quad \text{Ans: } 53$$

$$\sqrt{2809} = 53$$

$$(87)^2 \quad \begin{array}{l} \text{---} 100 = 13 \\ \text{---} 13 \\ \hline 74 \end{array}$$

$$(13)^2 = 169$$

$$\begin{array}{r} 74 \\ 169 \\ \hline 7569 \end{array}$$

$$(87)^2 = 7569$$

$$(48)^2 \quad \begin{array}{l} \text{---} 25 = 23 \\ \text{---} 50 = 2^2 = 4 \end{array}$$

$$(48)^2 = 234$$

$$(193 \times 53) - 7569 = x^2 + 234 - 5$$

$$10229 - 7569 = x^2 + 229$$

$$2660 - 229 = x^2$$

$$2431 = x^2$$

$$\boxed{x = 49}$$

18.

$$\sqrt{(569 + \sqrt{32 + \sqrt{(255 + \sqrt{1156})})})}$$

$$\begin{array}{r} \sqrt{1156} \\ \underline{3} \nearrow 4 \\ 12 > 11 6 \end{array}$$

$$\sqrt{1156} = 34$$

$$\sqrt{289} = 17$$

$$\begin{array}{r} 34 \\ 255 \\ \underline{289} \end{array} \quad \begin{array}{r} 32 \\ 17 \\ \underline{49} \end{array}$$

$$\sqrt{49} = 7$$

$$\begin{array}{r} 569 \\ 7 \\ \underline{576} \end{array}$$

$$\sqrt{576} = 24$$

cube and cube roots :

Number

cube unit digit

1

1

4

4

5

5

6

6

9

9

0

0

2

8

8

 $2 \leftrightarrow 8$

2

3

 $3 \leftrightarrow 7$

7

7

3

key properties :

i. unit digit value.

ii. Odd / Even rule.

odd $\rightarrow$ cubes $\rightarrow$ Odd

Even $\rightarrow$ cubes $\rightarrow$ Even

iii. sign rule.

positive $\rightarrow$ cube $\rightarrow$ positive

Negative $\rightarrow$ cube $\rightarrow$ Negative.

iv. No. of digits

If n number has d digits, n^3

has roughly $3d-2$ to $3d$ digits.

v. sum of cubes.

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

vi. Difference pattern.

Difference between consecutive cubes

grows as $3n^2 + 3n + 1$

19.

$$\sqrt[3]{175616}$$

$$\sqrt[3]{175616} \rightarrow 175 \quad \checkmark \text{ (6)}$$

Ans: 56

$$\checkmark \text{ (5)}^3 = 125 < 175$$

20.

$$\sqrt[3]{287496}$$

$$\sqrt[3]{287496} \rightarrow 287 \quad \checkmark \text{ (6)}$$

Ans: 66

$$\checkmark \text{ (6)}^3 = 216$$

21.

Expand the cube $(31)^3$

$$(31)^3 \Rightarrow a=3 \quad b=1$$

a^3

$3a^2b$

$3ab^2$

b^3

$(3)^3$

$3(3)^2(1)$

$3(3)(1)^2$

$(1)^3$

27

$+2$

$[27]$

9

1

29

7

9

1

Ans: 29791

22. Find the smallest number when multiplied with 137592 will make the perfect cube product. Find the cube root.

137592 $\Rightarrow$ prime factorization

2		137592		3		5733
		68796				1911
2		34398		3		637
2		17199		7		91
3		5733		7		13

$$137592 \Rightarrow 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 7 \times 7 \times 13$$

To make the perfect cube, we need

$$13 \times 13 \times 7$$

$$\text{Then } \Rightarrow 2^3 \times 3^3 \times 7^3 \times 13^3$$

If we take cube root, then

$$2 \times 3 \times 7 \times 13 = 546$$

23. Find the value of $\sqrt[3]{1000} + \sqrt[3]{0.008} +$

$$\sqrt[3]{0.125}$$

$$\sqrt[3]{1000} = \sqrt[3]{(10)^3} = 10$$

$$\sqrt[3]{0.008} = \sqrt[3]{8 \times 10^{-3}} = \sqrt[3]{(2 \times 10^{-1})^3} = 0.2$$

$$\sqrt[3]{0.125} = \sqrt[3]{125 \times 10^{-3}} = \sqrt[3]{(5 \times 10^{-1})^3} = 0.5$$

$$10 + 0.2 + 0.5 = 10.7$$

Ans : 10.7

24. What is the Smallest number by which 243000 must be divided so that the quotient is a perfect cube?

Ex: $x^5 \times y^3 \times z^7$ is eqn.

then
$$\frac{x^{(5)} \times y^3 \times z^{(7)}}{x^2 \times z^1}$$

The number after the multiple of 3

The values must be less than the powers.

$$5 - 3 = 2 \text{ for } x$$

$$7 - 6 = 1 \text{ for } z$$

$$(5 < 3, 7 < 6) \quad \checkmark$$

$$(6 > 5, 9 > 7) \quad \times$$

Prime factorization $\Rightarrow$ 243000

$$\begin{array}{r|l} 243 \Rightarrow & 3 \mid 243 \\ & 3 \mid 81 \\ & 3 \mid 27 \\ & 3 \mid 9 \\ & 3 \mid 3 \\ & \underline{1} \end{array}$$

$$1000 \Rightarrow 10 \times 10 \times 10$$

$$10 \Rightarrow 2 \times 5$$

$$243000 \Rightarrow \frac{3^5 \times 2^3 \times 5^3}{3^2} \quad (3 < 5)$$

Ans : 9

Vedic maths

$$\times 5 = \times 10 / 2$$

$$\times 2 = \times 10 / 5$$

$$\times 25 = \times 100 / 4$$

$$\times 4 = \times 100 / 25$$

$$\frac{820}{25} = \frac{820 \times 4}{100} = \frac{3280}{100} = 32.8$$

Subtracting from 10, 100, 1000, 10000

$$1000 - 745 \Rightarrow$$

Ans : 255

$$\begin{array}{r} 9 \quad 9 \quad 10 \\ - \quad 7 \quad 4 \quad 5 \\ \hline 2 \quad 5 \quad 5 \\ \hline \end{array}$$

Multiplication of number 11-19

$$12 \times 17$$

Base = 10

$$12 - 10 \Rightarrow +2$$

$$17 - 10 \Rightarrow +7$$

$$\begin{array}{cc} 12 & +2 \\ & \swarrow \searrow \\ 17 & +7 \end{array}$$

$$12 + 7 = \textcircled{19}$$

$$17 + 2 = 19$$

$$2 \times 7 = \textcircled{14}$$

19 14

Multiply any number only by 11 :

$$25 \times 11 \Rightarrow$$

$$\begin{array}{ccc} & 25 & \\ \swarrow & & \searrow \\ 2 & (2+5) & 5 \end{array}$$

Ans: 275

$$436 \times 11 \Rightarrow$$

$$\begin{array}{ccccc} & 4 & 3 & 6 & \\ & \swarrow & & \searrow & \\ 4 & (4+3) & (3+6) & & 6 \end{array}$$

Ans: 4796

Multiply any number by 99, 999, 9999

$$72 \times 99$$

$$99 = 100 - 1$$

$$72 \times 100 \Rightarrow$$

$$7200$$

$$\begin{array}{r} 7200 \\ - 72 \\ \hline 7128 \end{array}$$

Squaring a number ending with 5 :

Multiply the first digit on left
with (itself + 1) next number and
put 25 on the end.

$$(45)^2 = 4 \times 5 = (20)$$

$$(5)^2 = (25)$$

Ans: 2025

25. Evaluate $105 \times 106 = ?$

$$105 - 100 = +5$$

$$106 - 100 = +6$$

$$\begin{array}{cc} 105 & +5 \\ & \swarrow \searrow \\ 106 & +6 \end{array}$$

$$105 + 6 = 111$$

$$106 + 5 = 111$$

$$5 \times 6 = 30$$

$$\text{Ans : } 111\ 30$$

26. Evaluate $99 \times 97 = ?$

$$99 - 100 = -1$$

$$97 - 100 = -3$$

$$\begin{array}{cc} 99 & -1 \\ & \swarrow \searrow \\ 97 & -3 \end{array}$$

$$99 - 3 = 96$$

$$97 - 1 = 96$$

$$\text{Ans : } 9603$$

$$-3 \times -1 = 03$$

27. Evaluate 1397×1397

$$(1400 - 3)^2 = 1960000 + 9 - 8400$$

$$\text{Ans : } 1951609$$

28. Evaluate: $(112)^2$

$$\begin{aligned}(112)^2 &\Rightarrow \text{Base} = 100 \xrightarrow{\text{diff}} (12) \\ 112 + (12) &= (124) \checkmark \\ (12)^2 &= (144) \checkmark\end{aligned}$$

Ans: 112 144

29. Evaluate: $(1004)^2$

$$\begin{aligned}(1004)^2 &= (1000 + 4)^2 \\ &= (10^3)^2 + (4)^2 + 2(10^3)(4) \\ &= 1000000 + 16 + 8000 \\ &= 1008016 \text{ „}\end{aligned}$$

30. Evaluate: $(103 \times 105) + (432 \times 11) -$

$$(999 \times 47) = ?$$

$$103 \times 105$$

$$103 - 100 = +3$$

$$105 - 100 = +5$$

$$\begin{array}{cc} 103 & +3 \\ & \swarrow \searrow \\ 105 & +5 \end{array}$$

$$103 + 5 = 108$$

$$105 + 3 = 108$$

$$3 \times 5 = 15$$

$$103 \times 105 = 10815$$

$$432 \times 11$$

$$\begin{array}{cccc} & 4 & 3 & 2 \\ / & & & \backslash \\ 4 & (4+3) & (3+2) & 2 \end{array}$$

$$432 \times 11 = 4752$$

$$47 \times 999$$

$$\begin{array}{r} 47000 \\ - \quad 47 \\ \hline 46953 \end{array}$$

$$47 \times 999 = 46953$$

$$10815 + 4752 - 46953 = -31386$$

Multiplying the numbers with same
digits :

$$\underline{46} \times \underline{44}$$

Same tens digit.

$$\text{Tens digit} = 4 \times 5 \text{ (Next digit)}$$

$$\text{Tens digit} = \textcircled{20} \checkmark$$

$$\text{Ones digit} = 6 \times 4 = \textcircled{24} \checkmark$$

$$\text{Ans : } 2024$$

31. Multiply 59×51

$$\underline{59} \times \underline{51}$$

$$5 \times 6 = \textcircled{30}$$

$$9 \times 1 = \textcircled{09}$$

$$\text{Ans : } 3009$$

32. Multiply 63×67

$$\underline{63} \times \underline{67}$$

$$6 \times 7 = \textcircled{42}$$

$$3 \times 7 = \textcircled{21}$$

Ans: 4221

Multiply by 12 :

$$27 \times 12$$

$$27 \times 10 = 270$$

$$27 \times 2 = 54$$

$$\begin{array}{r} \textcircled{1} \\ 270 \\ + 54 \\ \hline 324 \end{array}$$

Ans: 324

33. Multiply 175×12

$$175 \times 10 = 1750$$

$$175 \times 2 = 350$$

$$\begin{array}{r} 1750 \\ + 350 \\ \hline 2100 \end{array}$$

Ans: 2100

34. Multiply 103×108

Base = 100

$$103 \quad +3$$

$$103 + 3 = 111$$

$$108 \quad +8$$

$$108 + 3 = \textcircled{111}$$

$$3 \times 8 = \textcircled{24}$$

Ans: 11124

35. Multiply 97×98

Base = 100

$$97 \quad -3$$

$$97 - 2 = 95$$

$$98 \quad -2$$

$$98 - 3 = \textcircled{95}$$

$$-3 \times -2 = \textcircled{6}$$

Ans: 9506

36. Multiply 103×97

Base = 100

$$103 \quad +3 \quad 103 - 3 = 100$$

$$97 \quad -3 \quad 97 + 3 = 100$$

$$+3 \times -3 = -9$$

$$97 + 3 = 100 \quad \downarrow$$

$$\begin{array}{r} 100 \ 00 \\ -3 \times +3 = - \quad 9 \\ \hline 9991 \end{array} \quad \times \text{Base } 100$$

Ans: 9991

37. Multiply 205×207

Base = 200

$$205 \quad +5 \quad 205 + 7 = 212$$

$$207 \quad +7 \quad 207 + 5 = 212$$

$$+5 \times +7 = 35$$

BUT THE REAL TWIST,

Since Base = 200

$$207 + 5 = 212 \times 2 = 424$$

If Base = 300 then multiply by 3

Ans: 42435

38. Multiply 1007 $\times$ 998

Base = 1000

$$1007 \quad +7$$

$$1007 - 2 = 1005$$

$$998 \quad -2$$

$$998 + 7 = 1005$$

$$+7 \times -2 = -14$$

$$\begin{array}{r} 1005000 \\ - \quad \quad 14 \\ \hline 1004986 \end{array}$$

Ans: 1004986

39. Multiply 23×99

$$23 \times 100 = 2300$$

$$23 \times 1 = 23$$

$$\begin{array}{r} 2300 \\ - \quad 23 \\ \hline 2277 \end{array}$$

Ans : 2277

40. Multiply 43×17

②

$$\begin{array}{r} 43 \times 17 \\ \hline 301 \\ 17 \\ \hline 471 \end{array}$$

Ans : 471

V - BODMAS

V - Vinculum

B - Brackets $() > \{\} > []$

O - order (of, exponent, squares, cubes, root)

D - division, M - Multiplication

Division and Multiplication have same precedence. Evaluate from Left to Right.

A - Addition, S - subtraction.

41. $72 \div 9$ of $2 \times 3 + 14 - (8 \div 4 \times 2)$

$$72 \div (9 \times 2) \times 3 + 14 - (2 \times 2)$$

$$72 \div 18 \times 3 + 14 - 4$$

$$4 \times 3 + 14 - 4$$

$$12 + 14 - 4$$

$$22$$

42. Evaluate $5.8 + [7.4 - \{3.1 \times (4.2 - 2.2)\}] \div 0.2$

$$5.8 + [7.4 - \{3.1 \times 2.0\}] \div 0.2$$

$$5.8 + [7.4 - 6.2] \div 0.2$$

$$5.8 + 1.2 \div 0.2$$

$$5.8 + 6$$

$$11.8$$

43. Evaluate $3^x + 3^{x+1} + 3^{x+2} = 351$

$$3^x [1 + 3 + 3^2] = 351$$

$$3^x \times 13 = 351$$

$$3^x = 351 / 13$$

$$3^x = 27$$

$$3^x = 3^3$$

$$\boxed{x=3}$$

44. Find $x^2 + y^2$ if $x + y = 21$ and $xy = 110$

$$(x + y)^2 = x^2 + y^2 + 2xy$$

$$(21)^2 = x^2 + y^2 + 2(110)$$

$$441 - 220 = x^2 + y^2$$

$$\boxed{x^2 + y^2 = 221}$$

45. Evaluate $(\frac{3}{5} \times (\frac{25}{18} \div \frac{5}{12})) + (\frac{7}{8} \times 16)$

$$- \sqrt{529} + (4^3 \div 2^4)$$

$$\left(\frac{3}{5} \times \left(\frac{\cancel{25}^5}{\cancel{18}_6 \cancel{3}} \times \frac{\cancel{12}^4 \cancel{2}}{\cancel{5}} \right) \right) + \left(\frac{7 \times \cancel{16}^2}{\cancel{8}} \right) - \sqrt{529}$$

$$+ \frac{\cancel{4} \times \cancel{4} \times 4}{2^2 \times \cancel{2}^2}$$

$$\frac{3}{5} \times \frac{10^2}{3} + 14 - 23 + 4$$

$$2 + 14 - 23 + 4$$

$$-3$$

46. A sum of RS. 1360 has been divided among A, B, C. A gets $\frac{2}{3}$ of what B gets. B gets $\frac{1}{4}$ of what C gets. What is the share of B.

A	B	C
$\frac{2}{3} (\frac{1}{4}x)$	$\frac{1}{4}x$	x
$\frac{1}{6}x$	$\frac{1}{4}x$	x

$$\frac{x}{6} : \frac{x}{4} : x = 1360$$

$$2x : 3x : 12x = 1360$$

$$17x = 1360$$

$$\boxed{x = 80}$$

47. To fill a tank, 25 buckets needed. If the bucket size is reduced to $\frac{2}{5}$ value then how many buckets are needed to fill the tank.

Let the bucket be 10 l, then the tank size is 250 l.

The bucket size is reduced to $\frac{2}{5}$ value

$$\text{then } x \rightarrow \frac{2}{5} x$$

$$x \rightarrow \frac{2}{5} \times 10 = 4 \text{ l}$$

$$\frac{250}{4} = 62.5 \text{ buckets.}$$

48. 8 people are planning to share equally the cost of a rental car. If one person withdraws from the arrangement and others share equally the entire cost of

the car. Then the each values increased by..

$$\text{original value} = \frac{C}{8}$$

$$\text{New value} = \frac{C}{7}$$

$$\text{New} - \text{OG} = \frac{C}{7} - \frac{C}{8} = \frac{C}{56}$$

$$\frac{\text{New} - \text{OG}}{\text{OG}} = \frac{C}{56} \times \frac{8}{C} = \frac{C}{7}$$

49. Find the Smallest Fractions.

$$\frac{1}{2}, \frac{3}{4}$$

$$\frac{1}{2} = 0.5$$

$$\frac{3}{4} = 0.75$$

$\frac{1}{2}$ is the Smallest Fractions.

50. Which Fraction is closed to 1 $\frac{19}{20}$,

$$\frac{29}{31}, \frac{49}{50} ?$$

$$\frac{19}{20} = 0.95$$

$$\frac{49}{50} = 0.98$$

$$\frac{29}{31} = 0.93$$

$\frac{49}{50}$ is the fraction closest to 1.

51. Evaluate $x - \frac{2}{5}x = 400$

$$5x - 2x = 400 \times 5$$

$$3x = 400 \times 5$$

$$x = 666.66$$

52. Sort in Ascending order

$$\frac{5}{6}, \frac{7}{12}, \frac{3}{4}, \frac{1}{2}$$

$$\frac{5}{6} = 0.83$$

$$\frac{3}{4} = 0.75$$

$$\frac{7}{12} = 0.58$$

$$\frac{1}{2} = 0.5$$

$$\frac{1}{2} < \frac{7}{12} < \frac{3}{4} < \frac{5}{6}$$

53. In a company there are $\frac{3}{8}$ are women,
men = ? Total employees are 640.
How many men are there?

$$\frac{3}{8} \Rightarrow \text{women}, \quad \frac{5}{8} \Rightarrow \text{men.}$$

$$\frac{5}{8} \times \overset{80}{640} = 400$$

$\therefore$ There are 400 mens in the company.

54. How much $\frac{7}{(8/9)}$ exceeds $\frac{(7/8)}{9}$

$$\frac{7}{8} \times 9 - \frac{7}{8 \times 9} = \frac{7}{8} \left[9 - \frac{1}{9} \right]$$

$$= \frac{7}{8} \times \frac{81-1}{9}$$

$$= \frac{7 \times 80}{8 \times 9} = \frac{70}{9}$$

55. Which Fraction is the Smallest ?

$$\frac{23}{28}, \frac{14}{15}, \frac{15}{19}, \frac{21}{24}$$

$$\frac{23}{28} = 0.82$$

$$\frac{14}{15} = 0.93$$

$$\frac{15}{19} = 0.79$$

$$\frac{21}{24} = 0.87$$

$\frac{15}{19}$ is the Smallest Fraction.

56. Evaluate $0.363 \times 0.522 + 0.363 \times 0.478 = ?$

$$0.363 (0.522 + 0.478)$$

$$0.363 (1)$$

$$0.363$$

57. $\frac{34.31 \times 0.473 \times 1.567}{0.0673 \times 23.25 \times 7.57}$ is closest

to 2.0, 2.15, 2.05, 1.15

$$\frac{34.31 \times 0.473 \times 1.567}{0.0673 \times 23.25 \times 7.57} = 2.15$$

58. Evaluate $\frac{(2.502 + 0.064)^2 - (2.502 - 0.064)^2}{2.502 \times 0.064}$

$$\frac{(a+b)^2 - (a-b)^2}{ab} = \frac{4ab}{ab} = 4$$

Ans: 4

59. Evaluate $\frac{4.5 \times 1.8 + 4.5 \times 8.2}{1.5 \times 4.5 + 1.5 \times 5.5}$

$$\frac{4.5}{1.5} \times \frac{1.8 + 8.2}{4.5 + 5.5}$$

$$\frac{45}{15} = 3$$

60. Evaluate $\frac{(0.02)^2 + (0.52)^2 + (0.035)^2}{(0.002)^2 + (0.052)^2 + (0.0035)^2}$

$$\frac{(0.02)^2 + (0.52)^2 + (0.035)^2}{(0.002)^2 + (0.052)^2 + (0.0035)^2}$$

$$\frac{(0.02)^2 + (0.52)^2 + (0.035)^2}{(0.02)^2 + (0.52)^2 + (0.035)^2} \times 100$$

Ans : 100

61. compare $3/5$ and $4/7$

$$3/5 = 0.6 \quad 4/7 = 0.5$$

$$\frac{3}{5} > \frac{4}{7}$$

62. Sort the sequence in ascending order.

$$\frac{1}{4}, \frac{2}{7}, \frac{3}{4}, \frac{4}{7}, \frac{5}{6}, \frac{6}{5}$$

$$\frac{1}{4} = 0.25 \quad \frac{3}{4} = 0.75 \quad \frac{5}{6} = 0.8$$

$$\frac{2}{7} = 0.28 \quad \frac{4}{7} = 0.57 \quad \frac{6}{5} = 1.2$$

$$\frac{1}{4} < \frac{2}{7} < \frac{4}{7} < \frac{3}{4} < \frac{5}{6} < \frac{6}{5}$$

63. Find the Fraction that is greater than $\frac{4}{5}$ and less than $\frac{6}{7}$

$$\frac{2}{3}, \frac{3}{4}, \frac{5}{6}, \frac{10}{11}$$

$$\frac{4}{5} = 0.8$$

$$\frac{6}{7} = 0.85$$

$$\frac{2}{3} = 0.6$$

$$\frac{3}{4} = 0.75$$

$$\frac{5}{6} = 0.82$$

$$\frac{10}{11} = 0.9$$

Ans: $\frac{4}{5} < \frac{5}{6} < \frac{6}{7}$

HCF - Highest Common Factor

$$12, 18$$

$$12 \rightarrow 2 \times 3 \times 2$$

$$2 \times 3 = 6$$

$$18 \rightarrow 2 \times 3 \times 3$$

HCF of 12, 18 is 6

64. Find the length of the plank which can be used to measure the length 4m 50cm, 9m 90cm, 16m 20cm in the highest time.

HCF: Largest, Highest, Dividing into equal.

LCM: Least, Minimum, Smallest, Repeating together, Repeat, Simultaneously, Traffic lights, Base lamps.

$$4\text{m } 50\text{cm} = 450\text{cm}, \quad 9\text{m } 90\text{cm} = 990\text{cm},$$

$$16\text{m } 20\text{cm} = 1620\text{cm}$$

$$450 = 10 \times 5 \times 3 = \underline{5} \times \underline{2} \times 5 \times \underline{3} \times \underline{3}$$

$$990 = 10 \times 11 \times 9 = \underline{5} \times \underline{2} \times \underline{3} \times \underline{3} \times 11$$

$$1620 = 10 \times 81 \times 2 = \underline{5} \times \underline{2} \times \underline{3} \times \underline{3} \times 2 \times 3 \times 3$$

$$\text{HCF} = 5 \times 2 \times 3 \times 3 = 90\text{cm}.$$

65. There are two numbers in the Ratio 5:11
and the HCF is 7. Find the Largest number.

$$\text{Ratio} = 5:11 \quad (A:B)$$

$$\text{HCF} = 7, \quad (H)$$

$$\text{Numbers} = AH, BH$$

$$\text{Numbers} \Rightarrow 5 \times 7 = 35, \quad 11 \times 7 = 77$$

$$\text{Ans: } 77$$

66. Find the LCM, HCF of the fractions

$$\frac{2}{3}, \frac{8}{9}, \frac{16}{81}$$

$$\text{LCM} = \frac{\text{LCM of Numerator}}{\text{HCF of Denominator}}$$

$$\text{HCF} = \frac{\text{HCF of Numerator}}{\text{LCM of Denominator}}$$

2, 8, 16

$$2 = 2$$

$$\text{HCF} = 2$$

$$8 = 2 \times 2 \times 2$$

$$\text{LCM} = 16$$

$$16 = 2 \times 2 \times 2 \times 2$$

3, 9, 81

$$3 = 3$$

$$\text{HCF} = 3$$

$$9 = 3 \times 3$$

$$\text{LCM} = 81$$

$$81 = 3 \times 3 \times 3 \times 3$$

$$\text{LCM} = \frac{16}{3}$$

$$\text{HCF} = \frac{2}{81}$$

Divisibles :

Divisible by 3: Add all the digits, the sum
is divisible by 3.

Divisible by 4: Last two digits are divisible by 4.

Divisible by 5: Unit digit must be 0 or 5.

Divisible by 7: Take the last digit, double it, subtract it with remaining number (Exclude 1's position).

$$53284 \rightarrow 4 \times 2 = 8$$

$$\begin{array}{r} 5328 \\ - 8 \\ \hline \end{array}$$

532 ⊖ → Exclude

$$53 \ominus \rightarrow 2 \times 2 = 4$$

$$53 - 4 = 49$$

$$49 \div 7 = 7$$

Divisible by 8: Last three digits are divisible by 8.

Divisible by 9: Sum of digits divisible by 9.

Divisible by 11: Sum of odd position -
Sum of Even position = Multiple of 11

Divisible by 12: Should be divisible by
both 3 and 4.

Divisible by 13: Multiply the last digit by 4,
add it to the remaining number (Exclude
1's position) Should be divisible by 13.

Divisible by 14: Divisible by both 7 and 2.

Divisible by 15: Divisible by both 3 and 5

Divisible by 16: Last 4 digits are divisible by 16.

Divisible by 17: Remove the last digit, multiply the
last digit by 5, Subtract from Remaining

number, Repeat if needed.

Divisible by 18: Divisible by both 9 and 2.

Divisible by 19: Remove the last digit, multiply the last digit by 2. Add to the remaining number repeat if needed.

Divisible by 20: Divisible by both 2 and 10.

67. If $43322K$ is exactly divisible by 72. What is K ?

$$\underline{43322K} = 22K$$

$$\begin{array}{r} 72 \\ / \quad \backslash \\ 8 \quad 9 \end{array}$$

$$\begin{array}{r} 28 \\ 8 \overline{) 22K} \\ \underline{16} \downarrow \\ 6K \\ \underline{64} \\ 0 \end{array} \quad \boxed{K=4}$$

$$4+3+3+2+2+K = \frac{18}{9} = 2$$

Ans: $K=4$

68. Which of the following numbers are divisible by 6? - $\underline{5406}$, $\underline{1753}$, $\underline{1717}$, $\underline{1655}$

$$\underline{5406} \quad \frac{06}{6} = 01 \quad \checkmark$$

Ans: 5406

69. If $785x3678y$ is divisible by 72.

Find $x-y$.

$$\begin{array}{r} 98 \\ 8 \overline{) 78y} \\ \underline{72} \\ 6y \\ \underline{64} \\ 0 \end{array}$$

$$\boxed{y=4}$$

$$\begin{array}{c} 72 \\ / \quad \backslash \\ 8 \quad 9 \end{array}$$

$$7+8+5+x+3+6+7+8+4$$

$$= \frac{48+x}{9} \quad \boxed{x=6}$$

$$x-y = 6-4$$

$$x-y = 2$$

70. Find the number that is divisible by 11.

12144, 21345, 54135, All of these.

$$12144 \quad (1+1+4) - (2+4)$$

$$6 - 6 = 0/11 = 0 \checkmark$$

$$21345 \quad (2+3+5) - (1+4)$$

$$10 - 5 = 5$$

$$54135 \quad (5+1+5) - (4+3)$$

$$11 - 7 = 4$$

Ans: 12144

71. Find the smallest number that should be added to 4725 to make it divisible by 8.

$$\begin{array}{r} 688 \\ 8 \overline{) 4725 + x} \\ \underline{424} \\ 52 \\ \underline{48} \\ 45 + 3 \\ \underline{48} \\ 0 \end{array}$$

$$\therefore x = 3$$

72. Find the missing digit x of $9724x$ that makes the number divisible by 9.

$$9 + 7 + 2 + 4 + x = 18 + 4 + x = \frac{22 + x}{9}$$

$$\boxed{x = 5}$$

73. Find the missing number of $97215x6$ that makes the number divisible by 11.

$$97215x6$$

$$(9 + 2 + 5 + 6) - (7 + 1 + x) = 22 - (8 + x) \\ = 22 - 11 = 11/11$$

$$\boxed{x = 3}$$

74. Find the missing value of $46490\#$ that makes the number divisible by 72

$$46490\#$$

$$\begin{array}{c} 72 \\ / \quad \backslash \\ 8 \quad 9 \end{array}$$

$$\begin{array}{r}
 113 \\
 8 \overline{) 90\#} \\
 \underline{8} \\
 10 \\
 \underline{8} \\
 2\# \\
 \underline{24} \\
 0
 \end{array}$$

$$\# = 4$$

$$4 + 6 + 4 + 9 + 0 + 4 = \frac{27}{9} = 3$$

$$\text{Ans: } \# = 4$$

75. Find the number that 7386071 divides.

$$4, 9, 3, 11$$

x

$$7386071$$

$$(7 + 8 + 0 + 1) - (3 + 6 + 7)$$

$$16 - 16 = \frac{0}{11} = 0 \checkmark$$

$$7 + 3 + 8 + 6 + 0 + 7 + 1 = 32$$

$$\text{Ans: } 11$$

76. A 3-digit number $4p3$ is added with another 3-digit 984 to give four digit number $13q7$ which is divisible by 11 then $(p - q) = ?$

$$\begin{array}{r} 4 \quad p \quad 3 \\ 9 \quad 8 \quad 4 \\ \hline 13 \quad 2 \quad 7 \end{array}$$

$$p + 8 = 9$$

$$1327 = (1+2) - (3+7)$$

$$p = 6 \quad q = 4 \quad p - q = 6 - 4 = 2$$

$$\therefore p - q = 2$$

77. Find the no. of 3 digit numbers that are divisible by 6.

$$999 - 100 = 899$$

$$\therefore \text{Pigeon-Hole principle} = \left\lfloor \frac{\text{pigeons} - 1}{\text{holes}} \right\rfloor + 1$$

$$= \left\lfloor \frac{899 - 1}{6} \right\rfloor + 1$$

$$= 149 + 1$$

Ans : 150

Simple Interest:

Formula: $SI = \frac{PTR}{100}$

P → Original amount

T → Time period in years

R → Rate of Interest per annum

A → Total amount ($P + SI$)

SI → Simple Interest.

78. If the principal amount is 40,000, the rate of interest is 8%. After 3 years what is the simple interest?

$$P = 40,000 \quad R = 8 \quad T = 3 \text{ years}$$

$$SI = \frac{PTR}{100} = \frac{40000 \times 8 \times 3}{100}$$

$$SI = 9600$$

79. If a sum of money at simple interest amount of 850 in 3 years then the amount rises to 900 in 4 years, the principal sum is

$$A_3 = p + 3 SI$$

$$A_4 = p + 4 SI$$

$$A_4 - A_3 = SI \text{ for one year.}$$

$$900 - 850 = 50$$

$$A_4 = p + 4 SI$$

$$900 = p + 4(50)$$

$$\boxed{p = 700}$$

80. If a person got a loan of interest 5% for 9 years and the simple interest is 4016.25. What would be his principal amount?

$$SI = \frac{PTR}{100}$$

$$4016.25 = \frac{9 \times 5 \times x}{100}$$

$$x = \frac{4016.25}{9 \times 5}$$

$$x = 8925$$